Pick any one pair of homologous sides: AB in the first figure answers to DE in the second. Newton's claim is that *every* homologous pair stands in one and the same ratio — so $AB/DE = BC/EF = CA/FD = k$, a single number k that measures how much larger the one figure is than the other. But the *areas* do not grow in this same ratio: they grow in its *duplicate ratio*, that is, in the ratio k^2. Thus if the larger figure's sides are twice those of the smaller ($k = 2$), its area is not twice but *four* times as great ($k^2 = 4$); triple the side, and the area is ninefold. This is why doubling a map's scale quadruples the ink, and why the homologous sides — highlighted here — govern both lengths and areas at once. *Q.E.D.*